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$$5^{2x} + 5^3 = 5^{x+2} + 5^{x+1}$$

$$\Leftrightarrow t^2 + 125 - 25t - 5t = t^2 - 30t + 125 \quad 5^x = t$$

$$D = 900 - 500 = 400$$

$$\frac{30 \pm \sqrt{400}}{2} = \frac{30 \pm 20}{2}$$

$$\Leftrightarrow 5^x = 25 \text{ of } 5^x = 5$$

$$\Leftrightarrow x = 2 \text{ of } x = 1$$

References